Garments Production Analysis Dashboard

1.

A.Project Overview

This project involves cleaning, transforming, and analysing raw data using **Excel** and creating an interactive **Power BI dashboard** to derive meaningful business insights.

B.Objective

The main objective is to demonstrate data pre-processing techniques using Excel and an interactive Power BI dashboard visualization to make informed decisions.

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2. Data Sources

- **Source Description and Timeline:** https://archive.ics.uci.edu/dataset/597/productivity+prediction+of+garment+employees
- **Domain**: Sales Analytics

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3. Objectives

A. Productivity Performance Analysis

Analyze whether production teams are achieving their productivity targets.

Business Questions:

- Which teams have the highest productivity?
- Which departments are meeting targets?
- Which teams require improvement?

B.Target vs Actual Productivity Analysis

Compare planned productivity with actual productivity.

Insights:

- Identify over-performing teams
- Detect productivity gaps

C.Department Performance Analysis

Compare productivity between departments.

Insights:

- Is sewing more productive than finishing?
- Which department needs process improvement?

D. . Workforce Efficiency Analysis

Understand the relationship between workforce size and productivity.

Insights:

- Which teams achieve better productivity with fewer workers?

E. **Overtime Impact Analysis**

Analyze whether overtime improves productivity

Insights:

- Does more overtime increase production?

F. **Idle Time Analysis**

Identify productivity losses caused by idle workers.

- Insights: Which teams have maximum idle time?

G.Incentive Impact Analysis

Analyze whether incentives improve productivity.

Insights: Do higher incentives produce better output?

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4. Attribute (Column /Features) Details:

Attribute Name	Data type	Description
Date	Date	Production date
Quarter	String(Categorical)	Quarter of the year
Department	String(Categorical)	Production department (Sewing / Finishing)
Day	String(Text)	Day of production
Team	Integer(Whole number)	Team identification number
Targeted_productivity	Integer(Decimal Number)	Expected productivity target
SMV	Integer(Decimal Number)	Standard Minute Value required for production
WIP	Integer(Whole Number)	Work-in-progress quantity
Over_time	Integer(Whole Number)	Extra working time in minutes
Incentive	Integer(Whole Number)	Additional employee incentive
Idle_time	Integer(Whole Number)	Time when workers were inactive
Idle_men	Integer(Whole Number)	Number of idle workers
No_of_style_change	Integer(Whole Number)	Number of production style changes
No_of_workers	Integer(Whole Number)	Total workers involved
Actual_productivity	Integer(Decimal Number)	Achieved productivity

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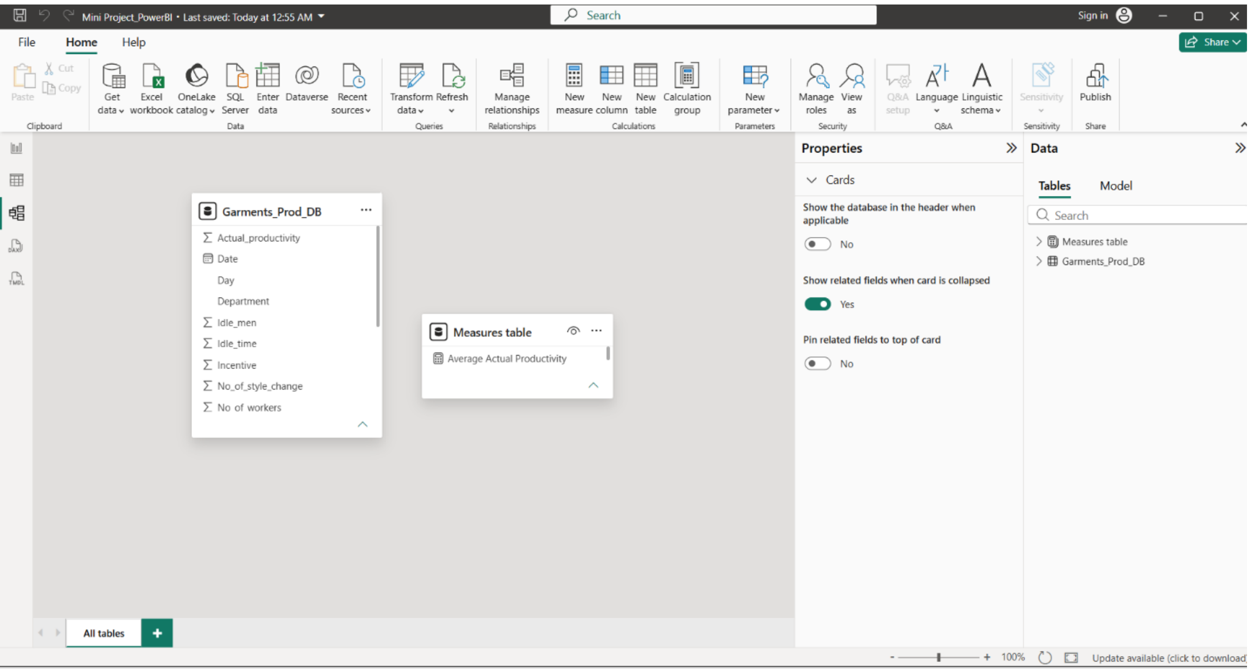

5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

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6. Data Pre-Processing (Excel / Power Query) Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.
- **Filtering & Sorting:** Organized data to focus on relevant records.
- **Pivot Tables:** Generated Pivot Tables for data summarization and initial insights.

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7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.

- **Calculated Columns & DAX Measures:**

Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals.

- **1. Total Production Records**
- **2. Average Actual Productivity**

3. Average Target Productivity

4. Productivity Gap

5. Productivity Achievement %

6. Total Overtime

7. Total Incentive

8. Average Workers

9. Total WIP

10. Average Idle Time

11. Highest Productivity

12. Lowest Productivity

13. Productivity Status

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8. Analysis and Visualizations (Power BI) Dashboard

Features:

- **Multiple Visualizations based on problem statement:** Bar charts, line charts, pie charts, cards, and tables to communicate insights.

  Charts/Visuals used:

  1.Stacked Column chart

  2.Stacked Bar chart

  3.Table

  4.Gauge

  5.Pie chart

  6.Scatter chart

  7.Line chart

  8.Ribbon chart

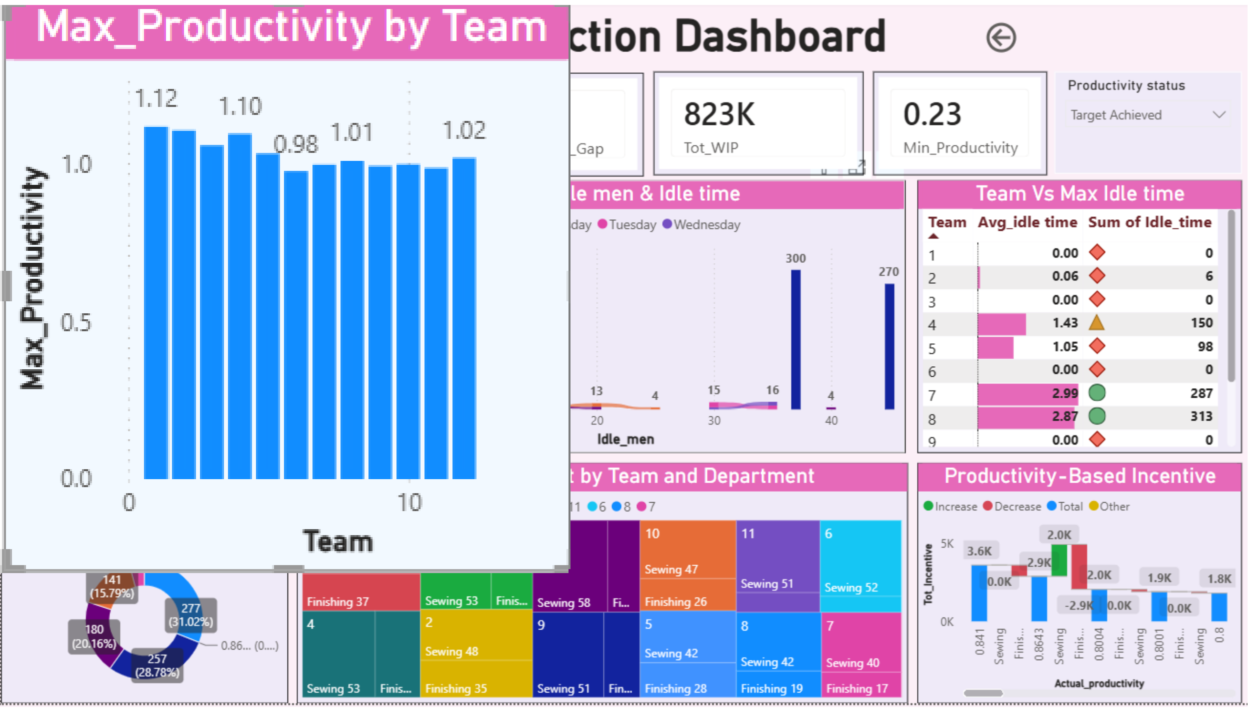
  9.Matrix

  10.Tree Map

  11.Waterfall chart

  12.Slicers

- **Consolidated Dashboard.**



9. Insights & Conclusions

Visual-1

Insight

 Sewing narrowly ***missed its productivity target by 1.36 units, achieving 99.7% of the target.***

 Finishing exceeded its productivity target by 7.89 units, achieving 102.1% of the target..

Analysis insights:

✦ **Descriptive**

✦ **Sewing achieved 99.7% of its target, missing by 1.36.**

✦ **Finishing exceeded its target by 7.89 (102.1% achievement).**

✦ **Diagnosis**

✦ Sewing experienced a slight productivity gap, while Finishing performed more efficiently than planned.

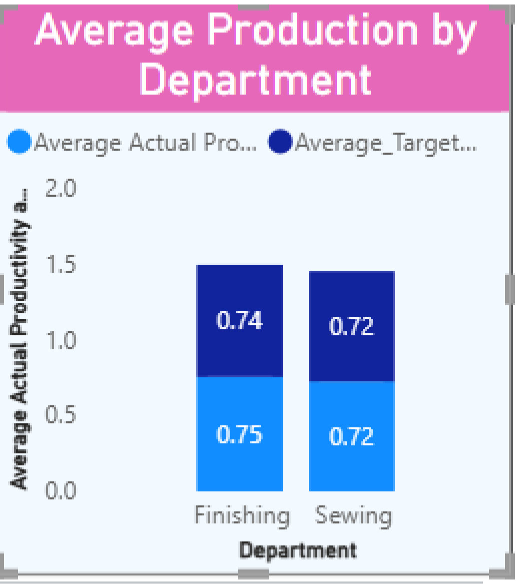
✦ **Predictive**

✦ Finishing is likely to continue exceeding targets, while Sewing is expected to remain close to target with minor improvements needed.

✦ **Prescriptive**

✦ Focus on improving Sewing efficiency and adopt Finishing's best practices to consistently meet or exceed productivity targets.

✦ **Visual 2**



Insight

- *Team 1 recorded the highest productivity (1.1204), leading all teams.*
- *Team 2 (1.1081) and Team 4 (1.0966) also performed above the benchmark of 1.0.*
- *Team 6 had the lowest productivity (0.9776), indicating the greatest improvement opportunity.*
- *Eight out of twelve teams achieved productivity at or above 1.0, reflecting generally strong performance.*
- *Productivity variation across teams is small, suggesting relatively consistent team performance overall.*

Descriptive

- **Team 1 achieved the highest productivity (1.1204), while Team 6 recorded the lowest (0.9776).**

Diagnostic

- **The performance gap suggests differences in team efficiency, resource utilization, or operational execution.**

Predictive

- **If current trends continue, Teams 1, 2, and 4 are likely to remain top performers, while Team 6 may continue to lag behind.**

Prescriptive

- **Analyze the best practices of high-performing teams and implement them in lower-performing teams, particularly Team 6, to improve overall productivity.**

Visual-3

Insight

Teams 2 and 8 achieved the highest production (109 units), while Team 11 recorded the lowest output (88 units), highlighting opportunities for performance improvement.

Descriptive

- **Teams 2 and 8 recorded the highest production (109 units), while Team 11 had the lowest production (88 units).**

Diagnostic

- **The production gap indicates variations in team efficiency, workforce utilization, or production capacity.**

Predictive Insight

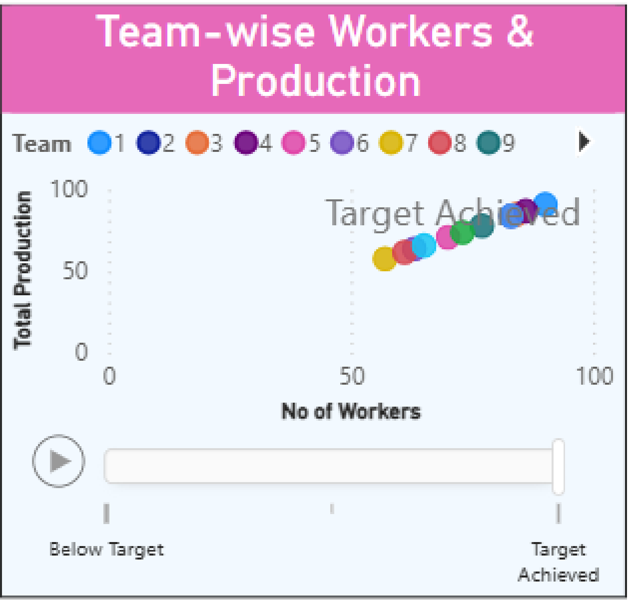
- **If current performance continues, Teams 2 and 8 are likely to remain the highest producers, while Team 11 may continue to underperform.**

Prescriptive Insight

- **Identify and replicate the best practices of Teams 2 and 8 to improve the production performance of lower-performing teams, especially Team 11.**

Visual-4

Insight



Finishing outperformed its target productivity (0.753 vs 0.737), while Sewing remained slightly below target (0.722 vs 0.724), indicating a need for minor efficiency improvements in Sewing.

- **Descriptive: Finishing achieved higher average productivity (0.753) and exceeded its target, while Sewing remained slightly below its target (0.722 vs 0.724).**
- **Diagnostic: Finishing's better performance indicates stronger process efficiency, whereas Sewing may have minor gaps in workflow or resource utilization.**
- **Predictive: Finishing is likely to maintain above-target productivity, while Sewing may require improvements to consistently meet targets.**
- **Prescriptive: Implement Finishing's best practices in Sewing and focus on improving operational efficiency to close the productivity gap.**

Visual-5

Insight

Actual productivity (880) exceeded the targeted productivity (873.37) by 6.63 units, indicating overall performance above expectations..

- **Descriptive:** Actual productivity exceeded the target by **6.63 units**, showing overall positive performance.
- **Diagnostic:** Higher productivity indicates effective operations, better resource utilization, and improved process efficiency.
- **Predictive:** Productivity is expected to remain above target if current performance trends continue.
- **Prescriptive:** Maintain current practices and identify key success factors to sustain and further improve productivity.

Visual -6

Insight

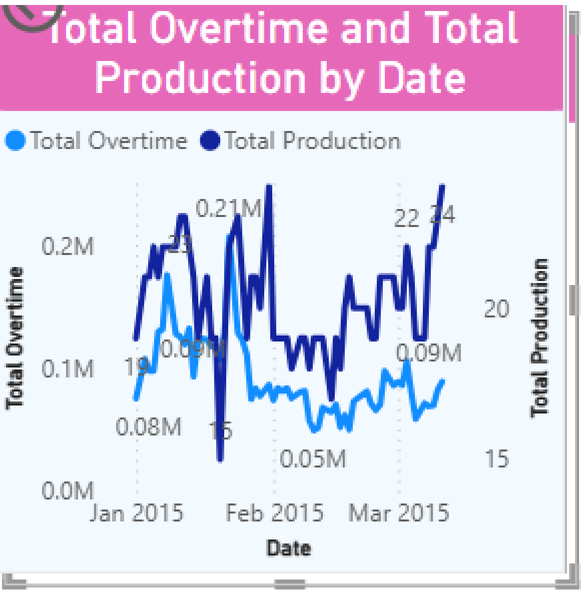
Most teams achieved their production targets with varying workforce levels, while below-target cases highlight opportunities to improve worker efficiency and productivity allocation across teams.

- **Descriptive:** All teams have a mix of target-achieving and below-target workers, showing variation in individual productivity performance.

- **Diagnostic:** Productivity gaps may be due to differences in worker efficiency, skills, workload distribution, or operational challenges.

- **Predictive:** Teams with more target-achieving workers are likely to maintain stronger production performance in the future.

- **Prescriptive:** Improve below-target worker performance through training, support, and by adopting practices from high-performing teams



Visual-7

Insight Sewing has the highest workforce strength with 691 workers compared to 506 workers in Finishing, indicating greater manpower allocation in the Sewing department.

 Descriptive: Sewing has a higher workforce count (**691 workers**) compared to Finishing (**506 workers**), indicating greater manpower allocation in Sewing.

 Diagnostic: The difference in workforce distribution may be due to higher production requirements, process complexity, or operational needs in Sewing.

 Predictive: If workforce allocation remains the same, Sewing is expected to continue handling a larger share of production activities.

 Prescriptive: Optimize workforce distribution based on productivity trends to ensure efficient utilization across both departments.

Visual-8

Insight Sewing holds the entire WIP inventory, with the highest WIP in Quarter 1 (333,380 units) and a declining trend across subsequent quarters, while Finishing maintains zero WIP throughout all quarters.

 Descriptive: Sewing holds the entire WIP inventory, with the highest WIP in Quarter 1 (333,380 units) and a continuous decline in later quarters; Finishing has zero WIP across all quarters.

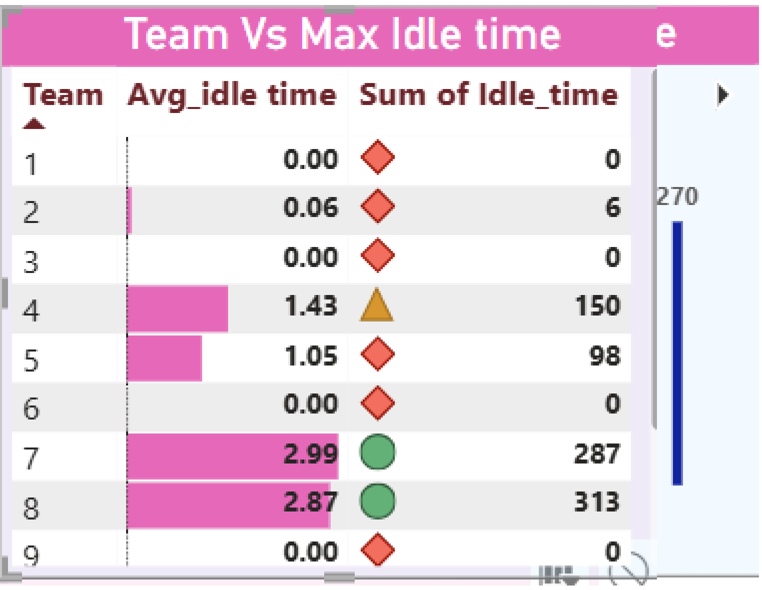
 Diagnostic: High WIP in Sewing indicates possible production bottlenecks, capacity imbalance, or delays before material moves to Finishing.

 Predictive: If the current trend continues, Sewing WIP is expected to reduce gradually, but accumulation risks may remain without process improvements.

 Prescriptive: Improve Sewing-to-Finishing workflow, identify bottlenecks, and optimize production flow to maintain balanced WIP levels.

Visual-9

Insight Production output shows fluctuations despite varying overtime levels, indicating that higher overtime does not always directly translate into higher production efficiency.



Team Vs Max Idle time

Team	Avg_idle time	Sum of Idle_time
1	0.00	0
2	0.06	6
3	0.00	0
4	1.43	150
5	1.05	98
6	0.00	0
7	2.99	287
8	2.87	313
9	0.00	0

- **Descriptive:** Overtime and production levels fluctuate across dates, with higher production not consistently matching higher overtime hours.

- **Diagnostic:** Variations indicate that overtime alone may not drive production; factors like workforce efficiency, workload, and process conditions may influence output.

- **Predictive:** Future production may continue to vary unless overtime utilization is optimized with better productivity planning.

- **Prescriptive:** Analyze overtime effectiveness and improve resource allocation to maximize production output without unnecessary overtime.

- **Visual-10**

Insight Idle time is highest on Saturday (up to 300 minutes) with more idle manpower, indicating potential resource utilization issues and opportunities to improve workforce allocation.

- **Descriptive:** Idle time is highest on Saturday (300 minutes) with significant idle manpower, while several days show minimal or zero idle time.

- **Diagnostic:** High idle time indicates possible workforce imbalance, scheduling issues, or inefficient resource utilization on specific days.

- **Predictive:** Without corrective actions, high idle periods may continue and impact overall operational efficiency.

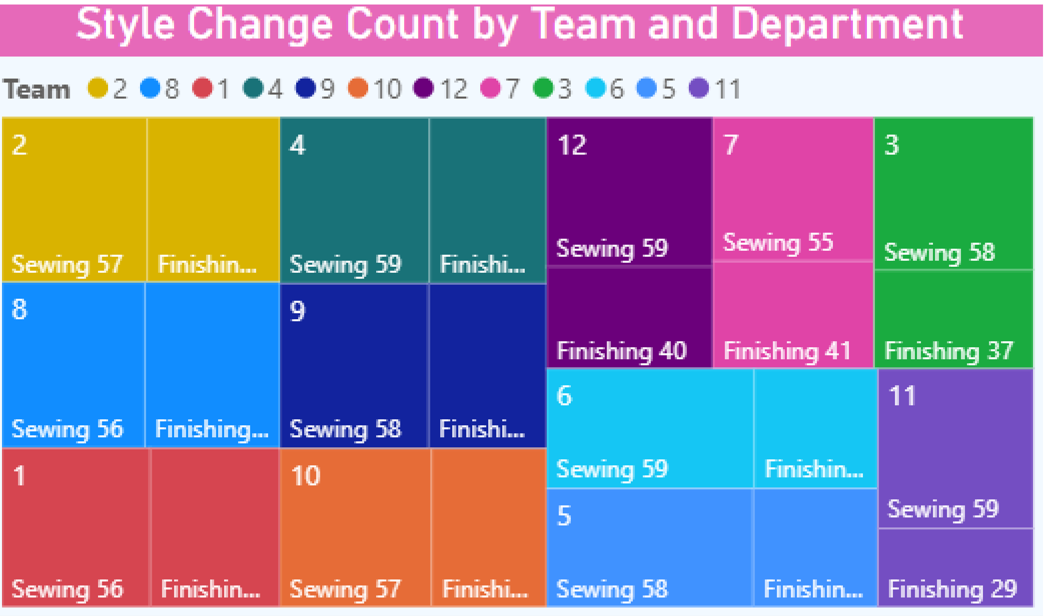
- **Prescriptive:** Optimize manpower planning, improve task allocation, and monitor high-idle days to reduce resource wastage.

Visual-11

Insight Teams 7 and 8 recorded the highest idle time (287 and 313 minutes), while Teams 1, 3, 6, 9, and 12 had zero idle time, indicating more efficient resource utilization..

Descriptive: Teams 7 and 8 recorded the highest idle time, while Teams 1, 3, 6, 9, and 12 maintained zero idle time, indicating better operational efficiency.

Diagnostic: High idle time in Teams 7 and 8 suggests possible workload imbalance, scheduling inefficiencies, or resource underutilization.



☐ **Predictive:** If current trends continue, Teams 7 and 8 may continue experiencing higher idle time, reducing overall productivity.

☐ **Prescriptive:** Optimize workload distribution and resource planning in Teams 7 and 8 to minimize idle time and improve operational efficiency.

Visual-12

Insight
Quarter 1 achieved the highest production (277) despite the highest average idle time (0.87), while the remaining quarters maintained lower or zero idle time with reduced production levels.

☐ **Descriptive:** Quarter 1 recorded the highest production (277) and the highest average idle time (0.87), while other quarters had lower production with minimal or zero idle time.

☐ **Diagnostic:** Higher idle time in Quarter 1 suggests production inefficiencies or temporary resource underutilization despite strong output.

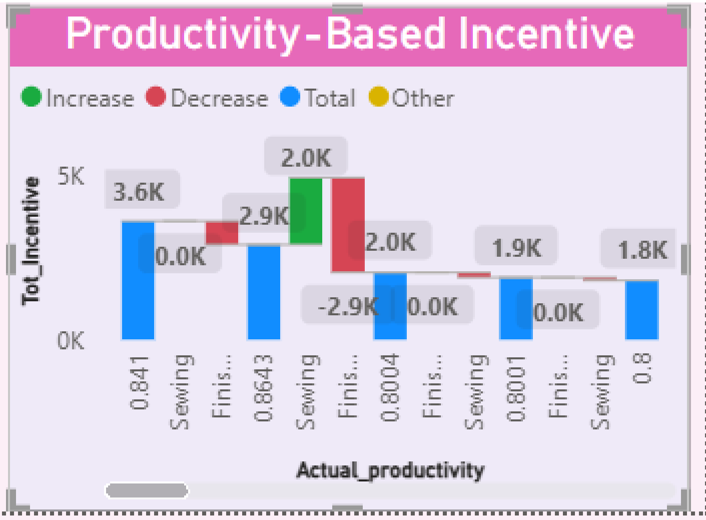
☐ **Predictive:** If current trends continue, future quarters may maintain lower idle time but could also experience reduced production levels.

☐ **Prescriptive:** Balance workforce utilization and production planning to sustain high output while minimizing idle time across all quarters.

Visual-13

Insight
Style changes were highest in Quarter 1 and Quarter 2 across both departments, with Sewing consistently recording more style changes than Finishing, while Quarter 5 had the fewest changes, indicating greater production stability..

- **Descriptive:** Style changes were highest in **Quarter 1 and Quarter 2**, with **Sewing consistently recording more style changes than Finishing** across all teams.
- **Diagnostic:** Frequent style changes, especially in Sewing, may have increased setup time and affected production efficiency.



- **Predictive:** If the current trend continues, Quarter 1 and Quarter 2 are likely to experience the highest operational disruptions due to style changes.
- **Prescriptive:** Reduce unnecessary style changes through better production planning, sequencing, and scheduling to improve overall efficiency.

Visual-14

Insight

Higher actual productivity generally corresponds to higher incentive payouts, with the highest incentives earned by high-performing employees in the Finishing department, while Sewing shows a wider distribution of productivity and incentive levels.

- **Descriptive:** Higher actual productivity generally resulted in higher incentive payouts, with the highest incentives awarded to top-performing employees.
- **Diagnostic:** Incentive variations reflect differences in productivity levels, indicating that rewards are closely linked to individual performance.
- **Predictive:** If productivity remains high, employees are likely to continue earning higher incentives under the current incentive structure.
- **Prescriptive:** Encourage productivity improvement through performance-based incentives, training, and sharing best practices across departments.

Final Dashboard Insights

Question: Which department has better productivity?
Answer:Finishing is the more productive and efficient department, while Sewing handles the higher production volume due to its larger workforce and production responsibilities.

Question: Which team exceeds production targets?

Answer: Team 2 is the strongest overall performer, consistently exceeding production targets through high productivity, the highest production output (tied), and minimal idle time. Team 1 and Team 4 are also top-performing teams.

Question: Does overtime improve productivity?

Answer: Overtime has a limited impact on productivity; improving workforce efficiency and process optimization is more effective for increasing production performance.

Question: How does workforce size affect output?

Answer: Increasing workforce size can improve production capacity, but optimal workforce allocation, skill utilization, and minimizing idle time are key factors in achieving higher productivity.

Question: Which teams have productivity issues?

Answer: Teams 7 and 11 require the highest attention due to productivity gaps, higher idle time, and lower output, while Teams 1, 2, and 4 are the benchmark teams for productivity performance.

Question: Does incentive impact performance?

Answer: Incentives positively support performance improvement, but maximizing productivity requires a combination of effective rewards, workforce utilization, and process efficiency.

Question: How much production loss occurs due to idle time?

Answer: Production loss due to idle time is mainly concentrated in high-idle teams (especially Teams 7 and 8); reducing idle time through better workforce planning can improve output without increasing manpower.

Question: How to improve production and over all efficiency?

Answer: Improving production efficiency requires reducing idle time, balancing department workloads, optimizing workforce utilization, and focusing on process improvements rather than relying on additional overtime or manpower.

10. **Conclusions**

The integration of Excel and Power BI proved effective for end-to-end data analysis, from raw data to visual reporting.